

```
In [1]: import sagemaker
from sagemaker import get_execution_role
```

```
/home/ec2-user/anaconda3/envs/python3/lib/python3.10/site-packages/pydantic/_internal/_fields.py:192:
UserWarning: Field name "json" in "MonitoringDatasetFormat" shadows an attribute in parent "Base"
warnings.warn(
```

```
[02/23/25 02:31:09] INFO Found credentials from IAM Role:
BaseNotebookInstanceEc2InstanceRole
```

```
sagemaker.config INFO - Not applying SDK defaults from location: /etc/xdg/sagemaker/config.yaml
```

```
sagemaker.config INFO - Not applying SDK defaults from location: /home/ec2-user/.config/sagemaker/con
fig.yaml
```

```
In [2]: # Get SageMaker execution role
role = get_execution_role()
```

```
[02/23/25 02:31:32] INFO Found credentials from IAM Role:
BaseNotebookInstanceEc2InstanceRole
```

```
In [3]: # Print the role to confirm
print("Execution Role:", role)
```

```
Execution Role: arn:aws:iam::156041433736:role/service-role/SageMaker-Kirsten
```

```
In [4]: # Create a SageMaker session
sagemaker_session = sagemaker.Session()
```

```
[02/23/25 02:31:59] INFO Found credentials from IAM Role:
BaseNotebookInstanceEc2InstanceRole
```

```
In [5]: # Upload data to S3
train_input = sagemaker_session.upload_data("data", bucket='flight-price-data-680', key_prefix="flight
```

```
In [6]: # Print the S3 location
print("Training data uploaded to S3:", train_input)
```

```
Training data uploaded to S3: s3://flight-price-data-680/flight-price-data
```

```
In [7]: from sagemaker.sklearn.estimator import SKLearn
```

```
script_path = 'flight_price_model.py' # Path to the training script
```

```
# Create an SKLearn Estimator
```

```
sklearn_estimator = SKLearn(
    entry_point=script_path,
    instance_type='ml.m5.xlarge', # Instance type for training
    framework_version="0.20.0", # Specify the framework version
    py_version="py3", # Python version
    role=role, # The role used for SageMaker
    sagemaker_session=sagemaker_session # SageMaker session
)
```

```
[02/23/25 02:36:07] INFO Found credentials from IAM Role:
BaseNotebookInstanceEc2InstanceRole
```

```
In [9]: # Start the training job
sklearn_estimator.fit({'train': train_input})
```

```
[02/23/25 02:40:29] INFO SageMaker Python SDK will collect telemetry to help us better tel
understand our user's needs, diagnose issues, and deliver
additional features.
To opt out of telemetry, please disable via TelemetryOptOut
parameter in SDK defaults config. For more information, refer
to
https://sagemaker.readthedocs.io/en/stable/overview.html#configuring-and-using-defaults-with-the-sagemaker-python-sdk.
```

```
INFO Creating training-job with name:
sagemaker-scikit-learn-2025-02-23-02-40-29-185
```

```

2025-02-23 02:40:30 Starting - Starting the training job...
..25-02-23 02:40:45 Starting - Preparing the instances for training.
..25-02-23 02:41:30 Downloading - Downloading the training image.
.2025-02-23 02:42:08,911 sagemaker-containers INFO      Imported framework sagemaker_sklearn_containe
r.training
2025-02-23 02:42:08,914 sagemaker-training-toolkit INFO      No GPUs detected (normal if no gpus insta
lled)
2025-02-23 02:42:08,923 sagemaker_sklearn_container.training INFO      Invoking user training script.
2025-02-23 02:42:09,115 sagemaker-training-toolkit INFO      No GPUs detected (normal if no gpus insta
lled)
2025-02-23 02:42:09,127 sagemaker-training-toolkit INFO      No GPUs detected (normal if no gpus insta
lled)
2025-02-23 02:42:09,138 sagemaker-training-toolkit INFO      No GPUs detected (normal if no gpus insta
lled)
2025-02-23 02:42:09,147 sagemaker-training-toolkit INFO      Invoking user script
Training Env:
{
  "additional_framework_parameters": {},
  "channel_input_dirs": {
    "train": "/opt/ml/input/data/train"
  },
  "current_host": "algo-1",
  "current_instance_group": "homogeneousCluster",
  "current_instance_group_hosts": [
    "algo-1"
  ],
  "current_instance_type": "ml.m5.xlarge",
  "distribution_hosts": [],
  "distribution_instance_groups": [],
  "framework_module": "sagemaker_sklearn_container.training:main",
  "hosts": [
    "algo-1"
  ],
  "hyperparameters": {},
  "input_config_dir": "/opt/ml/input/config",
  "input_data_config": {
    "train": {
      "TrainingInputMode": "File",
      "S3DistributionType": "FullyReplicated",
      "RecordWrapperType": "None"
    }
  },
  "input_dir": "/opt/ml/input",
  "instance_groups": [
    "homogeneousCluster"
  ],
  "instance_groups_dict": {
    "homogeneousCluster": {
      "instance_group_name": "homogeneousCluster",
      "instance_type": "ml.m5.xlarge",
      "hosts": [
        "algo-1"
      ]
    }
  },
  "is_hetero": false,
  "is_master": true,
  "is_modelparallel_enabled": null,
  "job_name": "sagemaker-scikit-learn-2025-02-23-02-40-29-185",
  "log_level": 20,
  "master_hostname": "algo-1",
  "model_dir": "/opt/ml/model",
  "module_dir": "s3://sagemaker-us-east-1-156041433736/sagemaker-scikit-learn-2025-02-23-02-40-29-1
85/source/sourcedir.tar.gz",
  "module_name": "flight_price_model",
  "network_interface_name": "eth0",
  "num_cpus": 4,
  "num_gpus": 0,
  "output_data_dir": "/opt/ml/output/data",
  "output_dir": "/opt/ml/output",
  "output_intermediate_dir": "/opt/ml/output/intermediate",
  "resource_config": {

```

```

    "current_host": "algo-1",
    "current_instance_type": "ml.m5.xlarge",
    "current_group_name": "homogeneousCluster",
    "hosts": [
        "algo-1"
    ],
    "instance_groups": [
        {
            "instance_group_name": "homogeneousCluster",
            "instance_type": "ml.m5.xlarge",
            "hosts": [
                "algo-1"
            ]
        }
    ],
    "network_interface_name": "eth0"
},
"user_entry_point": "flight_price_model.py"
}

Environment variables:
SM_HOSTS=["algo-1"]
SM_NETWORK_INTERFACE_NAME=eth0
SM_HPS={}
SM_USER_ENTRY_POINT=flight_price_model.py
SM_FRAMEWORK_PARAMS={}
SM_RESOURCE_CONFIG={"current_group_name":"homogeneousCluster","current_host":"algo-1","current_instance_type":"ml.m5.xlarge","hosts":["algo-1"],"instance_groups":[{"hosts":["algo-1"],"instance_group_name":"homogeneousCluster","instance_type":"ml.m5.xlarge"}],"network_interface_name":"eth0"}
SM_INPUT_DATA_CONFIG={"train":{"RecordWrapperType":"None","S3DistributionType":"FullyReplicated","TrainingInputMode":"File"}}
SM_OUTPUT_DATA_DIR=/opt/ml/output/data
SM_CHANNELS=["train"]
SM_CURRENT_HOST=algo-1
SM_CURRENT_INSTANCE_TYPE=ml.m5.xlarge
SM_CURRENT_INSTANCE_GROUP=homogeneousCluster
SM_CURRENT_INSTANCE_GROUP_HOSTS=["algo-1"]
SM_INSTANCE_GROUPS=["homogeneousCluster"]
SM_INSTANCE_GROUPS_DICT={"homogeneousCluster":{"hosts":["algo-1"],"instance_group_name":"homogeneousCluster","instance_type":"ml.m5.xlarge"}}
SM_DISTRIBUTION_INSTANCE_GROUPS=[]
SM_IS_HETERO=false
SM_MODULE_NAME=flight_price_model
SM_LOG_LEVEL=20
SM_FRAMEWORK_MODULE=sagemaker_sklearn_container.training:main
SM_INPUT_DIR=/opt/ml/input
SM_INPUT_CONFIG_DIR=/opt/ml/input/config
SM_OUTPUT_DIR=/opt/ml/output
SM_NUM_CPUS=4
SM_NUM_GPUS=0
SM_MODEL_DIR=/opt/ml/model
SM_MODULE_DIR=s3://sagemaker-us-east-1-156041433736/sagemaker-scikit-learn-2025-02-23-02-40-29-185/source/sourcedir.tar.gz
SM_TRAINING_ENV={"additional_framework_parameters":{},"channel_input_dirs":{"train":"/opt/ml/input/data/train"},"current_host":"algo-1","current_instance_group":"homogeneousCluster","current_instance_group_hosts":["algo-1"],"current_instance_type":"ml.m5.xlarge","distribution_hosts":[],"distribution_instance_groups":[],"framework_module":"sagemaker_sklearn_container.training:main","hosts":["algo-1"],"hyperparameters":{},"input_config_dir":"/opt/ml/input/config","input_data_config":{"train":{"RecordWrapperType":"None","S3DistributionType":"FullyReplicated","TrainingInputMode":"File"}}, "input_dir":"/opt/ml/input","instance_groups":["homogeneousCluster"],"instance_groups_dict":{"homogeneousCluster":{"hosts":["algo-1"],"instance_group_name":"homogeneousCluster","instance_type":"ml.m5.xlarge"}}, "is_hetero":false,"is_master":true,"is_model_parallel_enabled":null,"job_name":"sagemaker-scikit-learn-2025-02-23-02-40-29-185","log_level":20,"master_hostname":"algo-1","model_dir":"/opt/ml/model","module_dir":"s3://sagemaker-us-east-1-156041433736/sagemaker-scikit-learn-2025-02-23-02-40-29-185/source/sourcedir.tar.gz","module_name":"flight_price_model","network_interface_name":"eth0","num_cpus":4,"num_gpus":0,"output_data_dir":"/opt/ml/output/data","output_dir":"/opt/ml/output","output_intermediate_dir":"/opt/ml/output/intermediate","resource_config":{"current_group_name":"homogeneousCluster","current_host":"algo-1","current_instance_type":"ml.m5.xlarge","hosts":["algo-1"],"instance_group_hosts":[{"hosts":["algo-1"],"instance_group_name":"homogeneousCluster","instance_type":"ml.m5.xlarge"}],"network_interface_name":"eth0"},"user_entry_point":"flight_price_model.py"}
SM_USER_ARGS=[]
SM_OUTPUT_INTERMEDIATE_DIR=/opt/ml/output/intermediate
SM_CHANNEL_TRAIN=/opt/ml/input/data/train

```

```

PYTHONPATH=/opt/ml/code:/miniconda3/bin:/miniconda3/lib/python3.7.zip:/miniconda3/lib/python3.7:/miniconda3/lib/python3.7/lib-dynload:/miniconda3/lib/python3.7/site-packages
Invoking script with the following command:
/miniconda3/bin/python flight_price_model.py
2025-02-23 02:42:09,148 sagemaker-training-toolkit INFO      Exceptions not imported for SageMaker Debugger as it is not installed.
/miniconda3/lib/python3.7/site-packages/sklearn/externals/joblib/externals/cloudpickle/cloudpickle.py:47: DeprecationWarning: the imp module is deprecated in favour of importlib; see the module's documentation for alternative uses
    import imp
/miniconda3/lib/python3.7/site-packages/sklearn/utils/validation.py:37: DeprecationWarning: distutils Version classes are deprecated. Use packaging.version instead.
    LARGE_SPARSE_SUPPORTED = LooseVersion(scipy_version) >= '0.14.0'
/miniconda3/lib/python3.7/site-packages/sklearn/externals/joblib/numpy_pickle.py:104: DeprecationWarning: tostring() is deprecated. Use tobytes() instead.
    pickler.file_handle.write(chunk.tostring('C'))
/miniconda3/lib/python3.7/site-packages/sklearn/externals/joblib/numpy_pickle.py:104: DeprecationWarning: tostring() is deprecated. Use tobytes() instead.
    pickler.file_handle.write(chunk.tostring('C'))
/miniconda3/lib/python3.7/site-packages/sklearn/externals/joblib/numpy_pickle.py:104: DeprecationWarning: tostring() is deprecated. Use tobytes() instead.
    pickler.file_handle.write(chunk.tostring('C'))
2025-02-23 02:42:11,388 sagemaker-containers INFO      Reporting training SUCCESS

2025-02-23 02:42:29 Uploading - Uploading generated training model
2025-02-23 02:42:29 Completed - Training job completed
Training seconds: 79
Billable seconds: 79

```

```

In [10]: # Deploy the trained model to an endpoint
deployment = sklearn_estimator.deploy(
    initial_instance_count=1, # Number of instances to deploy
    instance_type="ml.m5.xlarge" # Instance type for inference
)

print("Model deployed successfully!")

```

```

[02/23/25 02:44:30] INFO      Creating model with name:
sagemaker-scikit-learn-2025-02-23-02-44-30-566
[02/23/25 02:44:31] INFO      Creating endpoint-config with name
sagemaker-scikit-learn-2025-02-23-02-44-30-566
INFO      Creating endpoint with name
sagemaker-scikit-learn-2025-02-23-02-44-30-566
-----!Model deployed successfully!

```

```

In [11]: deployment.endpoint

[02/23/25 02:49:03] WARNING    The endpoint attribute has been renamed in sagemaker>=2.
See: https://sagemaker.readthedocs.io/en/stable/v2.html for details.

```

```

Out[11]: 'sagemaker-scikit-learn-2025-02-23-02-44-30-566'

```

```

In [12]: # Use the deployed endpoint to make predictions

import json
import numpy as np
import boto3

# Create a runtime client
runtime_client = boto3.client('sagemaker-runtime')

# Example data for prediction (replace with actual feature values)
data = np.array([[0, 3, 2.5, 0]])

# Convert data to JSON for the endpoint
payload = json.dumps(data.tolist())

# Invoke the endpoint
response = runtime_client.invoke_endpoint(
    EndpointName=deployment.endpoint_name,
    ContentType='application/json',

```

```
    Body=payload
)

# Get the result from the response
prediction = json.loads(response['Body']).read().decode()
print("Prediction:", prediction)
```

Prediction: [4868.872972972973]

```
In [13]: # Delete the endpoint after use
         deployment.delete_endpoint()
```

[02/23/25 02:49:25] INFO Deleting endpoint configuration with name:
sagemaker-scikit-learn-2025-02-23-02-44-30-566

[02/23/25 02:49:26] INFO Deleting endpoint with name:
sagemaker-scikit-learn-2025-02-23-02-44-30-566